Institute Technology of Cambodia

Lab01 - Report

Info	Details
Course	Course
Author	Author
Instructor	Instructor
Date	10/22/2026

Misc Files

File: `communication-overhead.md`

````
# Communication overhead · clubhub-ITC-4G (v1.0, 10/21/2026)

## Paths

| Model                  | People n | Paths n(n-1)/2   | Minutes/week | Hours/wee
k |
| ---------------------- | -------- | ---------------- | ------------ | ---------
- |
| (a) everyone + client | 5        | 5(5-1)/2=10      | 10x10 = 100  | 1.666
|
| (b) client via PO     | 4        | 4(4-1)/2 + 1 = 7 | 7x10 = 70    | 1.1666
|

Assumption: 10 minutes per path per week, because each communication path requir
es time to exchange updates, clarify requirements, and resolve misunderstanding
s.
Team capacity: 4 members × 2 h = 8 h/week.
→ model (a) costs 1.66 / 8 x 100 = 20.83% of weekly team capacity.
→ model (b) costs 1.166 / 8 × 100 = 14.53% of weekly team capacity.

## Our paths (model b)
```mermaid
flowchart LR
    KeoVimeanrathbosba --- Client
    KeoVimeanrathbosba --- SongVansak
    KeoVimeanrathbosba --- LorHengleap
    KeoVimeanrathbosba --- SreuVisal
    SreuVisal --- SongVansak
    SreuVisal --- LorHengleap
    SongVansak --- LorHengleap
````

Two practices

1. 10-minute stand-up, three times a week. Each member briefly reports progress, blockers, and their next task. Bosba collects client-related questions and shares the answers with the team. This replaces scattered status updates with three short meetings, costing approximately 0.5 h/week in scheduled meeting time. We will keep updates brief and move detailed discussions to GitHub Issues.
````

2. GitHub Issues as the single source of truth. We record requirements, decisions, bugs, and task owners in GitHub Issues instead of repeatedly explaining the same information in private messages. This reduces the assumed cost per path from 10 to 6 minutes per week. Model (b) then costs 7 × 6 = 42 minutes, or 0.7 h/week. This saves approximately 0.47 h/week compared with the original model.

A sixth member in week 11

With six team members and the client communicating only through the PO:

Path n(n-1)/2 = 6(6-1)/2 + 1 = 16

The overhead rises from 7 to 16 paths, an increase of (16-7)/7 × 100 ≈ 128.6%. At 10 minutes per path, that is 160 minutes or 2.67 h/week.

Decision: We would not accept the sixth member in week 11 unless the additional person has a clearly defined role that provides enough benefit to justify the onboarding and coordination costs, consistent with Brooks's law.

Reflection: the essential difficulty we expect first

Reflection

The essential difficulty that our team is most likely to encounter first is changeability in ITC Club Hub. As we navigate ITC Club Hub, it is completely expected of us to get confused especially with the rapid transformation demands from the client. For demonstration, the rules for importing club data may initially require only a club name and capacity. However, the client may want to add other fields such as category, meeting schedules and so on. These abrupt requests of change can affect the CSV format, validation rules, C++ classes, and tests.

For the next four weeks, our team will be on guard and prepared with the rapid request of changes from the client by recording issues that lie in GitHub, while making sure we have the confirmations made by the Product Owner and the client. We will also work on our collaboration skills within our team ensuring that we communicate well from Product Owner to Scrum Masters to Developers. The quality of our work will improve as anticipated as we learn to work together.

Signed Members Statement

- Song Vansak (Scrum Master): I will organize the short team check-in, focusing more on communication, progress, GitHub issues.
- Lor Hengleap (Developer 1): I expect to find more bugs than planned especially when integrating , so i will test each feature before adding to the system.

- Sreu Visal (Developer 2): I will test my code regularly, read error messages carefully, and fix problems as soon as they appear.
- Keo Vimeanrathbosba (Product Owner): I will be checking in with the client to make sure we can give them what they ask for and what they really want. Additionally, I will work well with my team.

````
#### File: `decision.md`
```markdown

# Decision log · clubhub-ISE-4G (v0.1, 10/21/2026)

| #   | Date       | Decision            | Decided by   | Alternatives considere
d |
| --- | ---------- | ------------------- | ------------ | ----------------------
- |
| 1   | 10/14/2026 | Team name"ISE-4G"   | vote 4-0     | ISE
|
| 2   | 10/14/2026 | Scrum Master Vansak | ProductOwner | None
|
| 3   | 10/14/2026 | Working Agreements  | Everyone     | None
|
````

File: `failure-case-analysis.md`

````
# Failure case analysis ·Therac-25 (v1.0, 10/21/2026)

| Authors             | Section |
| ------------------- | ------- |
| Keo Vimeanrathbosba | 1-2     |
| Song Vansak         | 3-4     |
| Lor Hengleap        | 5-6     |
| Sreu Visal          | 7       |

## 1. Context

Therac-25 was a computer-controlled medical linear accelerator manufactured by A
tomic Energy of Canada Limited. Hospitals used the machine to treat cancer by di
recting high-energy electron or X-ray beams at tumors [2].
Medical staff entered the prescribed treatment mode, energy, and dose through a
computer terminal. Software then controlled and monitored the machine. Unlike th
e earlier Therac-6 and Therac-20 machines, Therac-25 depended more heavily on so
ftware for safety and did not retain several independent hardware interlocks
[2].

## 2. Timeline
````
````

Date	Event
22 March 1983	The U.S. Food and Drug Administration determined that Therac-25 was substantially equivalent to an existing radiation-therapy device through the 510(k) process [1].
3 June 1985	A patient at Kennestone Regional Oncology Center in Marietta, Georgia, received a massive radiation overdose during an electron treatment. The connection between the injury and Therac-25 was not initially accepted or thoroughly investigated [2].
26 July 1985	A patient at the Ontario Cancer Foundation in Hamilton, Ontario, received an overdose after the machine repeatedly displayed an “H-tilt” error and indicated that no dose had been delivered [2].
21 March 1986	A patient at the East Texas Cancer Center in Tyler, Texas, received a massive overdose. Investigators initially could not reproduce the malfunction [2].
11 April 1986	A second patient at the East Texas Cancer Center received an overdose. The hospital physicist and operator reproduced the sequence that triggered the failure, allowing the software problem to be investigated [2].
15 April 1986	Atomic Energy of Canada Limited submitted a medical-device report to the FDA covering the two Tyler accidents [2].
17 January 1987	Another overdose occurred at Yakima Valley Memorial Hospital. In February 1987, regulators required routine use of Therac-25 to stop until permanent corrective modifications were implemented [2].

3. What failed

Therac-25 depended heavily on software instead of independent hardware safety controls. Certain fast changes made by an operator could leave the software in an unsafe state. Unclear error messages then allowed the operator to continue treatment without knowing that the patient had received dangerous radiation [2].

```mermaid
flowchart TD
    A[Hardware safety controls were reduced] --> B[Safety depended on software]
    B --> C[Rapid editing triggered a software defect]
    C --> D[Machine entered an unsafe beam mode]
```

```
D ⟶ E[Error message did not explain the danger]
E ⟶ F[Patient received a massive overdose]
```

4. Root causes

Root cause	Type	Difficulty	Justification
Unsafe shared data and timing behavior	Technical	Complexity	Several software tasks interacted in ways that were difficult to predict [2].
Insufficient testing of rapid operator input	Process	Invisibility	The defect appeared only during a particular sequence and timing of inputs [2].
Independent hardware interlocks were reduced	Organisational	Invisibility	An independent safety layer could have stopped the unsafe radiation beam [2].

5. Practices that would have prevented or limited it

Root cause	Practice that could prevent or limit it	Course chapter/topic
Shared data and timing problems	Define safe state transitions; test rapid inputs and timing sequences	Chapter 01
Inadequate testing of rapid inputs	Add regression tests for fast edits, timeouts, and retries	Chapter 01
Operators couldn't see the unsafe state	Show the actual failure clearly and block unsafe continuation	Chapter 01

6. Lessons for ITC Club Hub

1. Don't record attendance when a student's session has expired; show a clear error instead.
2. Test rapid retries, timeouts, and repeated check-ins so they can't create an incorrect attendance record

7. Sources

1. Author/Organization: U.S. Food and Drug Administration
   Title: “510(k) Premarket Notification: Therac 25 Linear Accelerator (K837152)”
   Year: 1983
   URL: View the FDA Therac-25 record
   (https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?ID=k837152)

2. Authors: Nancy G. Leveson and Clark S. Turner
   Title: “An Investigation of the Therac-25 Accidents”
   Year: 1993
   URL: Read the Therac-25 investigation
   (https://web.mit.edu/6.033/2004/wwwdocs/papers/Therac_1.html)

````
#### File: `failure-risks.md`
```markdown
# Failure risks · ITC Club Hub (v0.1, 10/21/2026)

| #  |                    How the project could fail                | Chapter 01
topic | Early warning sign | First countermeasure | Owner |
|----|----------------------------------------------------------------------
-----------------------------------------------------|-----------------|-----
--------------|----------------------|-------|
| F1 | Inaccurate or failed attendance logging due to session/login timeouts and
authentication bugs during peak event check-in times. | Problem Statement | High
rate of user complaints about login failures and manual attendance override requ
ests after club meetings. | Implement robust session management, local error cac
hing, and fallback offline check-in options. | Lead Developer |
| F2 | Low adoption rate among students and club leaders who continue using lega
cy group chats and paper sheets. | Today's Workaround | Fewer than 20% of active
campus clubs creating accounts or uploading course/event materials within the fi
rst month. | Conduct hands-on onboarding sessions with club leaders and introduc
e single-tap QR check-in to prove convenience over paper. | Project Manager |
| F3 | Ambiguous positioning leading users and evaluators to view the system as
a generic file portal rather than an integrated hub. | Product Vision | Stakehol
ders ask how the platform differs from standard cloud drives or existing social
channels. | Refine and communicate the unique value proposition uniting material
s, directory, and automated attendance. | Product Owner |
| F4 | Scope creep caused by attempting to build full backend systems in C++ whi
le designing complex web/mobile UIs within a single semester. | Scope Definition
| Team missing early milestones or spending disproportionate time debugging C++
server bindings instead of core logic. | Strictly enforce the defined Scope boun
daries (C++ core backend/session logic only, web/mobile frontend UI modelled onl
y). | Technical Lead |
| F5 | Resistance or lack of support from lecturers and advisors due to privacy
concerns or misalignment with university policies. | Stakeholder Management | Sl
ow response times from faculty advisors when requested to review or pilot the pl
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```
atform. | Schedule early administrative briefing demos highlighting automated re
porting and data security controls. | Stakeholder Liaison |
```

File: `problem-statement.md`

```
# Problem statement · ITC Club Hub (v0.1, 10/21/2026)

## Problem

The problem of logging in affects students' experience in using ITC Club Hub;
the impact of which is inaccurate self-record attendance.
A successful solution would record students' attendances more accurately.

## Vision

For lecturers, students, or parents who do not know about us, ITC Club Hub is a
central platform, directory, or student union portal that manages and coordinate
s various student-led clubs, activities, and extracurricular initiatives.
Unlike a normal school app, our product provides many services for users such as
containing the materials for students including slides, labs, quizzes, and many
more.

## Today's workaround

Students register through chat groups, while organizers manually maintain partic
ipant lists in spreadsheets or messages.

## Scope

- Built in C++ this semester:
  * Core backend service logic or modular performance components (if required by
course specifications).
  * Secure credential hashing and token-based session management routines.
- Modelled only (web and mobile):
  * Interactive UI/UX wireframes for student dashboards, club directory pages, a
nd mobile attendance check-in interfaces.
  * Responsive screen designs for the course material repository (slides, labs,
and quizzes).
- Out of scope:
  * Automated facial recognition or biometric attendance verification.
  * Deep integration with external university registrar or official ERP/SIS data
bases.
  * Native mobile app store deployment (restricting mobile access to responsive
web interfaces).
```

File: `README.md`

```
# Lab 01 Record

## Files Produced

| File                                | Task
| Authors                   |
| ---------------------------------- | ----------------------------------------
----------- | ----------------------- |
| `README.md`                        | Repository information and team table
| Team                      |
| `.gitignore`                       | Ignore generated CMake build files
| Developer 1, Developer 2 |
| `LICENSE`                          | Temporary licence placeholder
| Team                      |
| `CMakeLists.txt`                   | Configure the C++20 project
| Developer 1, Developer 2 |
| `include/clubhub/banner.hpp`       | Declare the banner function and team name
| Developer 1, Developer 2 |
| `src/banner.cpp`                   | Build the version and team banner
| Developer 1, Developer 2 |
| `src/main.cpp`                     | Run and print the banner
| Developer 1, Developer 2 |
| `lab01/team-charter.md`            | Record team working agreements
| Team                      |
| `lab01/decision.md`               | Record team decisions
| Team                      |
| `lab01/failure-case-analysis.md`   | Analyse the Therac-25 failure
| Team                      |
| `lab01/problem-statement.md`       | Define the Club Hub problem and scope
| Product Owner             |
| `lab01/stakeholders.md`            | Identify Club Hub stakeholders
| Product Owner             |
| `lab01/failure-risks.md`           | Record five project failure risks
| Product Owner             |
| `lab01/README.md`                  | Record the files produced in Lab 01
| Developer 1, Developer 2 |
| `lab01/communication-overhead.md` | Compare communication paths and coordinati
on practices | Team                      |

## Inputs

Inputs reused from earlier labs: none (first lab)

## Change Log

| Date       | Version | Summary                                                              |
Commit
|
```

| ---------- | ------- | -- |

---------- |
| 2026-10-20 | v0.1.0 | Created the initial Club Hub C++20 and CMake project |
`Record Lab-01 Changes , LorHengleap liep.lor01@gmail.com , SreuVisal sreuvisal95@gmail.com` |

File: `stakeholders.md`

Stakeholders · ITC Club Hub (v0.1, 10/21/2026)

Stakeholder	Interest	Influence	Needs from Club Hub	How we reach them
★ Students	High	High	Easy login, seamless attendance recording, and quick access to club directory, slides, labs, and quizzes.	Telegram student group chats, campus club fairs, class representatives, university social media.
★ Club Leaders / Committee	High	High	Efficient member management, reliable attendance tracking tools, and centralized event & material announcements.	Direct messaging, club executive meetings, student council channels.
★ Lecturers / Course Instructors	High	High	Accurate, verifiable attendance reports, reduced administrative overhead, and oversight of academic club activities.	Faculty emails, departmental briefing sessions, direct faculty liaison.
Department Heads / Deans	Medium	High	High-level reports on student engagement, departmental activity alignment, and policy compliance.	Formal administrative memos, academic council meetings, official reports.
Campus IT Support Team	Medium	High	Secure system integration, low server overhead, compliance with campus IT protocols, and easy system maintenance.	Technical coordination meetings, support tickets, internal developer reviews.
University Administration	Medium	High	Standardized campus activity tracking, improved student engagement analytics, and official governance oversight.	Formal proposals, administrative meetings, pilot system demonstrations.
Parents	Low	Low	Visibility into student involvement, academic resources, and confirmation of safe campus activitie

```
s.                  | University public portals, parent-teacher newsletters, camp
us open days.                          |
```

File: `team-charter.md`

```
# Team charter · clubhub-ISE-4G (v1.0, 10/21/2026)

## Members and roles

| Member              | GitHub              | Role (Sprint 1) | Shared hat
|
| ------------------- | ------------------- | --------------- | ----------------
|
| Keo Vimeanrathbosba | @keovimeanrathbosba | Product Owner   | Business-Analyst
|
| Song Vansak         | songvansak19        | Scrum Master    | Organizer
|
| Lor Hengleap        | KageKishi           | Developer       | DevOps
|
| Sreu Visal          | visal-rg            | Developer       | Tester
|

Scrum Master rotation: Sprint 1 = Song Vansak, Sprint 2 = Lor Hengleap.
The Product Owner stays the same all semester.

## Working agreements

1. Each member updates their assigned GitHub Issue with progress, blockers, and
next steps before each scheduled stand-up.
2. Every pull request is reviewed by one other member within 24 h (Mon-Fri); the
author merges after approval. (Lor hengleap).
3. Work/Focus on your own branch instead of the main branch.
4. Have a meet up/meeting at least once every week.
5. All members must help each other explain group project and To-Do path.
6. At the end of each feature, the team reviews completed tasks and discusses wh
at can be improved.

## Communication channels

| Purpose    | Channel             | Expected response    |
| ---------- | ------------------- | -------------------- |
| Chat       | Telegram Group Chat | 1-2 hour             |
| Work items | GitHub Issues       | within 12h           |
| Files      | this repository     | within a day         |
| Meetings   | IRL , VC            | at least once a week |

## Decision rules
```

- Product questions:product → PO decides after hearing everyone;
- Technical questions:consensus within 24 h, else majority vote;
- Deadlock (no agreement after 24-48 hours):ask the Professor at the next lab session.
- Where decisions are recorded: lab01/decision.md

Agreed by

1. Lor Hengleap "LorHengleap 10/20/2026"
2. Song Vansak "SongVansak 10/20/2026"
3. Sreu Visal "SreuVisal 10/20/2026"
4. Keo Vimeanrathbosba "KeoVimeanrathbosba 10/20/2026"